



Republic of Rwanda
Ministry of Education



SWDFA501

FRONTEND APP. DEVELOPMENT WITH REACT.JS

Develop Frontend Application using React.JS

Competence

RQF Level: 5

Learning Hours



Credits: 11

Sector: ICT AND MULTIMEDIA

Trade: SOFTWARE DEVELOPMENT

Module Type: Specific

Curriculum: ICTSWD5001 TVET CERTIFICATE V IN SOFTWARE DEVELOPMENT

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Purpose statement	This specific module describes the skills, knowledge and attitude required to develop Frontend applications using React.JS. This module is intended to prepare students pursuing TVET Level 5 in software development. Upon completion of this module, the learner will be able to: Develop React.JS application, Apply Tailwind CSS framework, Develop Next JS Application, Apply Progressive Web Application, and Deploy React Application					
Learning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		20%	Formative assessment	20%	50%
	Practical work:		80%		80%	
	Group project and presentation	30%				
	Individual project /Work	50%				
			Summative Assessment			50%

Knowledge	Skills	Attitudes
• Description of React.JS components, props, state, and lifecycle methods. • Explaining Virtual DOM and its role in enhancing performance.	• Installing NodeJS and Node Package Manager (NPM) • Installing React tools and libraries • Setup Next.JS project • Applying React basics • Building responsive and dynamic user interfaces using React.js.	• Curiosity • Problem-Solving Orientation • Attention to Detail • Continuous Learning • Adaptability • Team work

	• Creating reusable and modular components for efficient code maintenance. • Applying Tailwind CSS in React.JS • Utilizing React Hooks for managing state and side effects. • Implementing client-side routing for a seamless user experience. • Applying React Router for handling navigation in a React application. • Performing state management using tools like Redux or Context API.	
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Elements of Competence and Performance Criteria	
Elements of competence	Performance criteria

1. Develop React.js application	1.1. ReactJS environment is properly prepared based on application requirements.
	1.2. React basics are correctly applied based on application requirements.
	1.3. UI navigation is correctly applied based on application requirements
	1.4. React hooks are correctly applied based on components structure
	1.5. Events handling are properly implemented based on best practices
	1.6. API integration is properly implemented based on user requirements
2. Apply Tailwind CSS framework	2.1. Tailwind utility classes are correctly Applied based on UI Design.
	2.2. Responsive design principles are properly applied according to the required design
	2.3. Tailwind styles for a unique look are accurately customized based on tailwind configuration

3. Develop Next.JS Application	3.1. Typescript basics are properly applied based on project requirements.
	3.2. Routing is properly implemented based on components.
	3.3. API is properly created based on RESTful design principles.
	3.4. Application is properly secured based on system requirements and security standards.
4.Apply Progressive Web Application	4.1. Responsiveness is correctly maintained in line with user requirements.
	4.2. Web app manifest for an installable app experience is properly configured based on user requirements.
	4.3. Service workers for caching offline capabilities are correctly implemented based on user's requirements.
5. Publish the application	5.1. Environment variables are correctly setup based on deployment environments.
	5.2. React application is properly deployed based on platform requirements.
	5.3. Custom domain is appropriately Set up based on deployed application.
Course content	

Learning outcomes	At the end of the module the learner will be able to: 1. Develop React.js application 2. Apply Tailwind CSS framework 3. Develop Next JS Application 4. Apply Progressive Web Application 5. Deploy React Application
Learning outcome 1: Develop React.js application.	Learning hours: 40
Indicative content	
• Preparation of React js environment ✓ Definition of key concepts  ReactJS  Component  JSX (JavaScript XML)  Props  State  Lifecycle Methods  Hooks  Virtual DOM  React Router  Redux ✓ Introduction  Uses of react  Features ✓ Installation of NodeJS and Node Package Manager (NPM) ✓ Creating React Application	

- ✓ Explore React project structure
- ✓ Installation of additional React tools and libraries (React Developer tools)
- **Applying React basics**
 - ✓ React components
 -  Class components
 -  Functional components
 - ✓ JSX (JavaScript XML)
 - ✓ Props (Properties)
 - ✓ Lifecycle Methods
 -  componentDidMount
 -  componentDidUpdate
 -  componentWillUnmount
- **Applying UI navigation**
 - ✓ Installing React Route
 - ✓ Configuring Routes
 - ✓ Basic React Navigation
 - ✓ Handling 404 Pages
 - ✓ Redirects
 - ✓ URL Parameters
 - ✓ Nested Routing
- **Applying React hooks**
 - ✓ Identifying hooks
 -  State Hooks
 -  Effect Hooks
 -  Context Hooks
 -  Ref Hooks
 -  Callback Hooks

- ✓ Hook selection and combination
- ✓ Optimizing Performance
- ✓ Handling Complex State Logic
- ✓ Managing Global State

-  Context API

-  Redux

-  MobX

-  Zustand

- **Implementation of Events handling**

- ✓ Description of Events

-  Types of events

-  Synthetic events

-  Event bubbling

- ✓ Debouncing and Throttling Events

- ✓ Using Controlled Components

- ✓ Passing Arguments to Event Handlers

-  Arrow Function (in JSX)

-  Bind Method

- ✓ Use Custom Hooks for Event Listeners

- ✓ Handling Events on Dynamic Lists

- **Implementation of API integration**

- ✓ Initial Setup and Planning

-  Describe API

-  Dependencies Installation (Axios)

- ✓ Organizing API Calls

-  Defining and Grouping API Calls

-  Handling Data Fetching and Responses



Error Handling



Asynchronous Handling and Concurrency

✓ Performing API Security and testing

Resources required for the learning outcome

Equipment

- Computer
- Projector

Materials

- Internet

Tools

- VS Code
- Browser
- Terminal

Facilitation techniques or Learning activity

- Demonstration and simulation
- Practical exercise
- Individual work
- Trainer-guided
- Group discussion

Formative assessment methods / (CAT)

- Written assessment
- Oral presentation
- Performance assessment

Learning outcome 2: Apply Tailwind CSS framework.

Learning hours: 10

Indicative content

- **Applying Tailwind utility classes**
 - ✓ Integrating of Tailwind CSS in React.JS
 -  Install Tailwind CSS
 -  Configuring Tailwind CSS
 - ✓ Using Utility-First Fundamentals
 - ✓ Handling Hover, Focus, and Other States
 - ✓ Animation and Transitions

- ✓ Flexbox and Grid
- ✓ Reusing Styles
- ✓ Adding Custom Styles
- ✓ Functions & Directives

- **Applying responsive design principles**

- ✓ Mobile-First Approach
- ✓ Flexible Grid Layouts
- ✓ Responsive Images and Media
- ✓ Media Queries and Breakpoints
- ✓ Typography and Readability
- ✓ Interactive Elements
- ✓ Testing and Iteration

- **Customization of tailwind styles**

- ✓ Extending the Default Theme
- ✓ Adding Custom Variants
- ✓ Custom Fonts and Typography
- ✓ Customizing Colors
- ✓ Plugins for Additional Functionality
- ✓ Custom Directives for Complex Designs
- ✓ Conditional Styles with JavaScript

Resources required for the indicative content

Equipment	▪ Computer
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	▪ Projector
Materials	▪ Internet
Tools	▪ VS Code ▪ Browser ▪ Terminal
Facilitation techniques or Learning activity	▪ Demonstration ▪ Group discussion ▪ Practical exercise ▪ Trainer guided
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments
Learning outcome 3: Develop NextJS Application Learning hours: 20	
Indicative content	
● Applying TypeScript basics ✓ Environment setup  Installing TypeScript  Configuring TypeScript ✓ Implementing interface of variables ✓ Handling functions in TypeScript ✓ Data Handling  API data validation  Form validation  Error handling and exceptions ● Setup NextJS project ✓ Preparation of Environment	

✓ Project Creation

✓ Initial Development

 Creating Pages and components

 Implementing search engine optimization (SEO)

 Styling

 Caching Strategies

- **Implementing Rendering Techniques**

✓ Static Site Generation (SSG)

✓ Server-Side Rendering(SSR)

✓ Incremental Static Regeneration (ISR)

✓ Client-Side Rendering (CSR)

- **Implementing routing**

✓ Description of key concepts

 File-system Based Routing

 Dynamic Routes

 Nested Routes

 Link Component

 Programmatic Navigation

 API Routes

 Catch-all Routes

✓ Linking Components

✓ Programmatic Navigation

✓ Dynamic Routes

✓ Query Parameters

- **Creation of API**

- ✓ Define the API Endpoint

- ✓ Handling Request Types

- ✓ Using Dynamic API Routes

- ✓ Testing your API

- **Securing the Application**

- ✓ Performing Client-Side Security

-  Client-side rendering (CSR) security

-  Cross-Origin Resource Sharing (CORS)

-  Session management

-  Third-party libraries (Auth0)

- ✓ Performing Server-Side Security

-  HTTPS enforcement

-  Server-side rendering (SSR) security

-  API routes security

-  Content Security Policy (CSP)

-  Authentication

- ✓ Performing General Security Measures

Resources required for the indicative content

	Equipment	▪ Computer ▪ Projector
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- Computer
- Projector

	Materials	▪ Internet
	Tools	▪ VS Code ▪ Browser ▪ Terminal ▪ Postman
	Facilitation techniques or Learning activity	▪ Demonstration ▪ Group discussion ▪ Practical exercise ▪ Trainer-guided
	Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments
Learning outcome 4: Apply Progressive Web Application		Learning hours: 20
Indicative content		
• Maintain Responsiveness ✓ Leverage Progressive Enhancement ✓ Prioritize Mobile-First Design ✓ Utilize Performance Optimization Techniques • Configuring web application manifest ✓ Creating and Configuring the Manifest File ✓ Referencing the Manifest in Your HTML ✓ Testing and Validation • Implementation of service workers ✓ Describe Service workers ✓ Registration and Installation ✓ Caching Strategy Implementation		

✓ Updating service worker

Resources required for the indicative content

Equipment	▪ Computer ▪ Projector
Materials	▪ Internet
Tools	▪ VS Code ▪ Browser ▪ Terminal
Facilitation techniques or Learning activity	▪ Demonstration ▪ Group discussion ▪ Practical exercise ▪ Trainer-guided
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments

Indicative content

- **Configuration of environment variables**

- ✓ Set variables (backend host, ftp host, ftp-user, ftp-pass ...)

- ✓ Setup storage environment

-  Platform-specific options

-  Separate .env files

- **Deploying React Application**

- ✓ Run Build Script in React Application

- ✓ Configure Deployment Platform (Vercel)

- ✓ Migrate the application files

- ✓ Test the Deployed Application

- **Setup custom Domain**

- ✓ Description of DNS

-  Translation of Domain Names to IP Addresses

-  Hierarchy and Structure

-  Name Resolution Process:

- ✓ Configure DNS and SSL Settings

- ✓ Testing and verification

Resources required for the indicative content

Equipment

- Computer

	▪ Projector
Materials	▪ Internet
Tools	▪ VS Code ▪ Browser ▪ Terminal ▪ Git
Facilitation techniques or Learning activity	▪ Demonstration ▪ Group discussion ▪ Practical exercise
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments

Integrated/Summative assessment (For specific module)

TechTrends Ltd, is a leading business Company specializing in electronic gadgets. It is located in Kigali City. The company currently doesn't have a website where clients can find information about the company. The company is planning to develop a responsive website that can be accessible on all devices. As a software developer you are tasked by TechTrends Ltd to use React JS and develop a website that will be having the home, about and contact us pages to help the clients to get information about the company in a user-friendly and attractive view.

The website should be hosted with “**www.yourregno.site**” as a domain name and it will be set in a way that it can be searchable over search engines. On about page should show location using google map API.

Note:

Your regno: should be replaced by the student **SDMS Registration Number**

This task should be completed 6 hours.

All Tools and materials should be provided.

Resources

Tools	• VS Code
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	• Browser • Terminal • React Developer Tools (Browser Extension) • GitHub Actions / GitLab CI/CD • Material-UI / Bootstrap • Node.js and npm • ReactJS
Equipment	• Computer
Materials/ Consumables	• Internet

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome1. Develop React application (40%)	React js environment is prepared	React Application is created			2
		React tools and libraries are installed.			2
	UI navigation is correctly applied	React Route are installed			3
		React Route are configured.			3
		URL Parameters are applied			2
	React hooks are correctly applied	Hooks are selected and combined			3
		Performance is optimized			3
		Global State is managed			3
	Events handling are properly implemented	Controlled Components are used			3
		Arguments to Event Handlers are passed			3
		Custom Hooks for Event Listeners are used			3
		Dependencies are Installed			5

	API integration is properly implemented	API Calls are organized			5
Learning outcome 2: Develop NextJS Application (20%)	Tailwind utility classes are correctly applied	Tailwind CSS is installed and configured			3
		Utility-First Fundamentals are used			3
		Animation and Transitions are applied.			3
	Responsive design principles are properly applied	Interactive Elements are applied.			3
		Flexible Grid Layouts are applied.			3
		Mobile-First Approach are applied.			5
Learning outcome 3: (20%)	Typescript basics are properly applied	TypeScript is installed and configured			1
		interface variables is implement			4
	Next.js project is appropriately set up	Engine optimization (SEO) is implemented			3
		Project is created			2
	Routing is properly implemented	Components are linked			1
		Programmatic Navigation are implemented			2
		Dynamic Routes and Query Parameters are implemented			1
	API is properly created based	API Requests are handled			2
		Dynamic API Routes are used			1
		API is tested			1
	Security is properly implemented	Client-side rendering (CSR) security is implemented			2
Learning Outcome 4.	Web app manifest is properly added	The Manifest File is created and configured.			1
		Manifest is Referenced in HTML			2

Apply Progressive Web Application (10%)	Responsiveness is correctly maintained	Performance Optimization Techniques are utilized.			2
		Progressive Enhancement are Leveraged			1
		Mobile-First Design are Prioritized			1
	Service workers correctly implemented	Registration and Installation are implemented			1
		Caching Strategy are implemented			1
		service worker are updated			1
Learning Outcome 5: Deploy application (10%)	Environment variables are configured	variables are Identified			1
		storage method is chosen			1
	React application is properly deployed	Run Build Script in React Application			1
		Deployment Settings are Configured			2
		the Application is Deployed			2
	Custom domain is appropriately Set up	DNS and SSL Settings are configured			2
		Testing and verification are performed			1
	Total marks		100		
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

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